

on the quantum machine

1. run

run the circuit
with the dials
where they are

$$|\psi(\theta)\rangle = \hat{U}(\theta)|+\rangle^{\otimes L}$$

2. measure

read the magnets,
many thousands
of times over

$$m = \pm 1 \text{ at } \langle \psi | \hat{\Pi}_{\pm} | \psi \rangle$$

on an ordinary computer

3. average

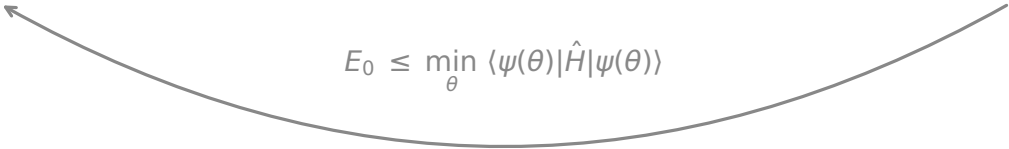
turn those
readings into
one energy

$$\hat{E} = \sum_{\alpha} c_{\alpha} \bar{P}_{\alpha}$$

4. decide

move the dials,
or stop if they
have stopped
helping

$$\theta \leftarrow \theta - \eta \nabla_{\theta} E$$


$$E_0 \leq \min_{\theta} \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle$$

and round again, until the dials stop improving

— whatever it reports is at or above the true lowest energy, so lower is always better